SOFTWARE REQUIREMENTS SPECIFICATION

MindBridge

Multimodal Intelligence Learning Platform

A web application that presents middle-school curriculum concepts through configurable learning pathways and evaluates learning outcomes.

Item	Details
Programme	Pinnacle Programme, IT Department
Team	Group of four students
Document	Software Requirements Specification (SRS)
Version	1.1 — Revised Draft
Date	10 October 2026
Status	For team/mentor review

Revision note: This version resolves the previously identified scope, scoring, learning-gain, adaptation, content approval, role, data, and testing gaps. Decisions that still require team confirmation are marked as open items.

Document Control

Version	Date	Author	Description
0.1	07-Oct-2026	Project team	Initial draft
1.0	07-Oct-2026	Project team	Full SRS draft
1.1	10-Oct-2026	Project team	Revised scope and requirements; clarified scoring, baseline testing, adaptation, roles, privacy, and validation

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1. Introduction

1.1 Purpose

This document specifies the functional, non-functional, and technical requirements for MindBridge / MI-Learn. It is intended for the four-person project team, Pinnacle mentors, and participating school representatives.

1.2 Problem Statement

Students may engage with curriculum concepts through different representations, such as language, reasoning, visualizations, sound, interactive activities, and real-world observation. The application will let students explore a learning preference profile and receive curriculum content through a recommended pathway. It will measure learning performance separately from preference and use that evidence to guide content recommendations.

1.3 Scope

In scope	Out of scope for current version
Middle-school students, primarily Classes 6–8	Primary and high-school grades
Six pathways: Linguistic, Logical-Mathematical, Spatial/Visual, Musical, Bodily-Kinesthetic, Naturalistic	Interpersonal and Intrapersonal pathways
Tamil Nadu State Board curriculum for selected classes/subjects	Native mobile application
Student, teacher, administrator roles; parent/guardian access only if approved for MVP	Payments, subscriptions, live video classes
Preference profile, pathway-based content, concept quizzes, progress reports	Automated AI publishing without teacher review

1.4 Definitions

Term	Meaning
Learning preference profile	Relative scores across the six supported learning pathways; not a diagnosis.
Primary pathway	Pathway with the highest normalized preference score.
Secondary pathway	Pathway with the second-highest normalized preference score.
Concept	A defined curriculum learning unit, e.g., Photosynthesis.
Baseline assessment	Concept-level assessment taken before the learning resource.
Learning gain	Normalized change between comparable concept pre-test and post-test scores.
Engagement indicators	Completion, time on task, retries, interaction and optional student feedback.
Mastery	Meeting the configured post-assessment criterion for a concept.

1.5 Conceptual framing

The system must not present its output as a clinical, psychological, or scientifically validated intelligence diagnosis. Use terms such as learning preference, learning pathway, engagement, and profile. A preference score must not be treated as proof that a student learns best through that pathway.

1.6 Background and references

The team reports background research on Multiple Intelligences, review of Tamil Nadu textbooks, and visits to one Government Middle School and one private school. Specific findings from the visits must be documented before the SRS is baselined. Online questionnaires may inform original question design, but should not be described as validated instruments unless independent evidence establishes that status.

2. Overall Description

2.1 Product perspective

The product is a responsive web application with connected assessment, learning, and evaluation modules. Teacher-managed curriculum resources are stored separately from application code.

2.2 Product workflow

```
Student registration/login
        ↓
Learning preference assessment
        ↓
Six-pathway profile (primary + secondary)
        ↓
Select curriculum concept
        ↓
Concept baseline quiz (when learning gain is measured)
        ↓
Teacher-approved pathway-based content
        ↓
Concept post-quiz + engagement feedback
        ↓
Learning gain / mastery analysis
        ↓
Continue primary pathway OR consider blended pathway
```

2.3 User classes and permissions

Role	Permitted actions
Student	Complete assessment, view own profile, learn, take quizzes, view own progress and provide feedback.
Teacher	View assigned classes/students, upload or draft resources, review/approve resources if granted permission, record class-test marks, view reports and provide observations.
Administrator	Manage accounts, roles, schools/classes, subjects, chapters, concepts, assessment configuration and system settings.
Parent/guardian (deferred unless approved)	If enabled, view only linked child's permitted progress and submit optional observations.
Content creator (permission, not necessarily separate role)	Create and tag draft resources; publishing requires an authorized approver.

2.4 Operating environment

- Modern versions of Chrome, Edge, or Firefox.
- Responsive use on school computers and tablets; mobile-browser support is desirable.
- Internet connection required for the MVP; optimize media for limited bandwidth.
- Hosting may be a college server or cloud service, with HTTPS for deployment.

2.5 Constraints and assumptions

- Content must align with the selected school curriculum and learning objectives.
- Students are minors; collect only necessary information and use appropriate school/parent consent for pilots.
- Teachers will provide or approve content and may enter classroom test results.
- The initial release will prioritize a small, complete workflow over broad content coverage.
- Tamil/English support, final subject scope, and hosting environment must be confirmed.

3. Specific Requirements

3.1 Supported learning pathways

Pathway	Representative formats
Linguistic	Stories, explanations, reading passages, vocabulary and analogies.
Logical-Mathematical	Patterns, puzzles, reasoning, cause-effect and structured problems.
Spatial/Visual	Diagrams, images, maps, animations, graphs and visualizations.
Musical	Rhymes, songs, audio explanations and rhythm-based memory aids.
Bodily-Kinesthetic	Interactive activities, simulations, drag-and-drop and demonstrations.
Naturalistic	Observation, classification, environmental examples and nature-based activities.

3.2 Module 1 — Pre-Assessment

3.2.1 Description

After first registration, the student is directed to a learning preference assessment. The module generates normalized scores for all six supported pathways, displays the profile, and records primary and secondary pathways. The profile can be retaken under defined rules.

3.2.2 Question types

- Preference/self-report questions: students choose formats they would prefer; there is no right or wrong answer.
- Scenario questions: students select how they would approach a learning situation; each option maps to one or more pathways.
- Interactive tasks: optional objectively scored tasks may measure task performance, but their correctness must not automatically be interpreted as preference evidence.
- Teacher/parent observations: optional supplementary input; must be identified separately from student-reported preference.

3.2.3 Scoring rules

Preference scores shall be calculated primarily from selected options and their pathway mappings. Each question has a configurable importance weight. Response time is recorded as a minor engagement/response-behaviour signal and must not directly penalize a student for answering slowly. Objective task performance is stored separately unless a documented, justified scoring rule is approved.

```
For each pathway k:
RawPreference(k) = Σ [option contribution(k) × question importance]

NormalizedPreference(k) = 100 × RawPreference(k) / MaximumPossiblePreference(k)

Primary pathway   = highest normalized preference score
Secondary pathway = second-highest normalized preference score
```

The implementation must define how unanswered items, multiple mappings, repeated attempts, and zero denominators are handled. A deterministic tie rule must be documented. If the top two scores are within a configurable near-tie margin, the profile may be presented as blended rather than implying a meaningful difference.

3.2.4 Functional requirements

ID	Requirement	Priority
PRE-F1	The system shall support student registration and login.	High
PRE-F2	A newly registered student shall be directed to the preference assessment before personalized learning.	High
PRE-F3	Each question option shall map to one or more of the six supported pathways.	High
PRE-F4	The assessment shall support preference, scenario-based, and optional interactive questions.	High
PRE-F5	The system shall save each response and response time.	High
PRE-F6	The backend shall aggregate pathway preference contributions using configured mappings and question weights.	High
PRE-F7	The system shall normalize all six pathway scores to a 0–100 scale, handling missing responses safely.	High
PRE-F8	The system shall rank six pathways and identify primary and secondary pathways using a documented tie/near-tie rule.	High
PRE-F9	The system shall display the six scores using an accessible chart and text summary.	Medium
PRE-F10	The system may collect optional teacher/parent observations, stored separately from student responses.	Low
PRE-F11	The system shall store profile history and make the current profile available to content recommendation.	High
PRE-F12	Authorized administrators shall manage questions, mappings, and importance weights.	High
PRE-F13	The system shall allow reassessment under defined rules without silently overwriting historical profiles.	Medium

3.2.5 Non-functional and technical requirements

- Questions and instructions must use age-appropriate language.
- The next question should load within two seconds under normal test conditions.
- Responses should be saved incrementally and recover safely after a connection interruption.
- The assessment should target a duration of 20–30 minutes, subject to pilot usability testing.
- The timer must not reset on refresh; server timestamps should be used where appropriate.
- Question coverage across all six pathways must be sufficient; equal item counts are not mandatory if normalization accounts for differences.
- Student responses and profiles must be accessible only to authorized users.

3.3 Module 2 — Specialized Content Learning

3.3.1 Description

The module presents a curriculum concept using resources associated with the student's primary pathway. The learning objective and core curriculum content must remain consistent across pathways; representation and activity may vary.

3.3.2 Curriculum hierarchy

```
School → Class/Grade → Subject → Chapter/Unit → Concept → Learning resources
```

3.3.3 Example: Photosynthesis

Pathway	Example resource
Linguistic	Simple explanation or story with key terms.
Logical-Mathematical	Flowchart and cause-effect reasoning about inputs and outputs.
Spatial/Visual	Animated diagram of sunlight, water and carbon dioxide producing glucose and oxygen.
Musical	Teacher-reviewed rhyme, song or audio mnemonic.
Bodily-Kinesthetic	Drag-and-drop simulation arranging process inputs and outputs.
Naturalistic	Guided observation of plants and discussion of growth conditions.

3.3.4 Concept and resource requirements

- Each concept shall include a learning objective, curriculum reference, difficulty level, and one or more resources.
- Each resource shall be tagged with grade, subject, chapter, concept, pathway, format, language, version, creator, and approval state.
- A concept shall have a short quiz or assessment linked to its learning objectives.
- The system shall record concept status: not started, in progress, completed, or needs review.
- Students may resume a concept without losing progress.
- The content-selection service shall use the current primary pathway by default and support a secondary/blended pathway when recommended or manually selected.

3.3.5 Content approval workflow

```
Draft / Uploaded → Pending Review → Approved → Published
                                 ■ Rejected / Needs Revision → Draft
```

Only approved and published resources may be shown to students. AI-assisted resources, if added, must be marked as drafts and reviewed by an authorized teacher before publication. The system shall retain approval status, reviewer, timestamp, and resource version.

3.3.6 Functional requirements

ID	Requirement	Priority
LRN-F1	The student shall browse grade, subject, chapter and concept.	High
LRN-F2	The system shall recommend approved content for the current primary pathway.	High
LRN-F3	The system shall support content for all six in-scope pathways.	High
LRN-F4	Content authoring/management shall be separate from the student learning interface.	High
LRN-F5	Authorized users shall upload, tag, edit, version and archive resources.	High
LRN-F6	The system shall offer secondary/blended pathway content when recommended or manually selected.	High
LRN-F7	The system shall support appropriate text, image, audio, video and interactive formats.	Medium
LRN-F8	Students shall be able to mark a concept complete when its completion conditions are met.	Medium
LRN-F9	The system shall save progress and allow the student to resume.	Medium
LRN-F10	A concept assessment shall be available after learning content.	High
LRN-F11	Student-visible resources shall require approved/published status.	High

3.3.7 Non-functional requirements

- Content must match the selected curriculum and be reviewed for age appropriateness.
- Media should be optimized for slow school networks; provide text alternatives where feasible.
- Images require alt text; audio/video should include transcripts or captions where feasible.
- Adding subjects and resources should not require changes to application code.
- Content and metadata should be stored separately from application logic.

3.4 Module 3 — Post-Assessment and Adaptation

3.4.1 Assessment sequence

When learning gain is measured for a concept, the system shall use a concept-level baseline assessment before the specialized content and a comparable post-assessment after it. The preference questionnaire is not an academic baseline test.

```
Concept → Baseline quiz → Learning content → Post-quiz → Analysis
```

3.4.2 Recorded data

- Baseline and post-test scores, quiz version, question responses, attempts and time taken.
- Concept completion and pathway/resource version used.
- Student feedback and engagement indicators, where collected.
- Teacher-recorded classroom assessment marks, with date and subject/concept when known.
- Profile version used at the time of learning, so later profile changes do not rewrite historical evidence.

3.4.3 Learning gain and mastery

For comparable pre-test and post-test scores on a 0–100 scale, normalized learning gain may be calculated as:

```
Learning Gain = (Post-test score − Pre-test score) / (100 − Pre-test score)
```

Example: pre-test 40, post-test 70 gives (70–40)/(100–40) = 0.50, or 50% normalized gain. If the pre-test score is 100, the denominator is zero; the system shall mark normalized gain as not applicable and report the raw score change instead. Negative gain is valid and must not be silently clamped to zero.

Mastery is a separate decision based on a configurable concept pass mark and/or rubric. Improvement does not automatically mean mastery. Pre/post assessments should cover the same learning objectives and have comparable difficulty.

3.4.4 Adaptation logic

```
Post-assessment result
        ↓
Check completion, learning time, quiz alignment, attempts and difficulty
        ↓
Sufficient evidence of low learning?
    No → Continue/offer support and gather more evidence
    Yes → Recommend primary + secondary blended pathway
        ↓
Re-teach/retry and reassess
        ↓
Record pathway used and outcome
```

The system shall not automatically switch pathways after a single poor score without checking whether the student completed the content and whether the assessment was suitable. Student or teacher override shall be available. The threshold for mastery and any adaptation rule must be configurable and validated during pilot testing.

3.4.5 Functional requirements

ID	Requirement	Priority
PST-F1	The system shall provide a concept quiz after learning.	High
PST-F2	The system shall record quiz score, attempts, responses and time.	High
PST-F3	Authorized teachers shall be able to enter classroom test scores.	High
PST-F4	The system shall calculate concept mastery using a configured criterion.	High
PST-F5	The system shall calculate learning gain only when comparable concept pre/post scores exist.	High
PST-F6	The system shall record engagement indicators separately from academic scores.	Medium
PST-F7	The system shall evaluate adaptation conditions using documented rules and thresholds.	High
PST-F8	Satisfactory learning shall allow the student to continue to the next concept.	High
PST-F9	When evidence supports adaptation, the system shall recommend secondary/blended content and permit reassessment.	High
PST-F10	Students shall view their own progress in accessible charts and text.	Medium
PST-F11	Teachers shall view authorized student and class reports.	Medium
PST-F12	Students may provide optional feedback on content helpfulness/difficulty.	Low
PST-F13	The system shall keep a history of scores, resource versions, and pathway changes.	High

4. System-Wide Requirements

4.1 User interface

- Child-friendly and responsive screens with clear navigation and large controls.
- Separate views for student, teacher, and administrator; parent access is deferred unless approved for the MVP.
- Charts must have labels or textual alternatives; do not rely on colour alone.
- Support keyboard navigation, readable contrast, alt text, and captions/transcripts where feasible.

4.2 Security and privacy

- Use role-based authorization on both frontend and backend; hiding a button is not sufficient access control.
- Passwords must be stored using a modern password-hashing algorithm; do not store plaintext or reversibly encrypted passwords.
- Use HTTPS in deployment and secure session/token handling, including expiration and logout.
- Students may access only their own records; teachers may access only assigned classes/students; administrators receive only the access needed for their duties.
- Collect the minimum necessary student data and establish school/parent consent for pilot testing.
- Define backup, restoration, retention, and deletion procedures for student records.
- Do not expose student profiles publicly or use student data for unrelated purposes.

4.3 Reliability and performance

- Save assessment and quiz submissions reliably; prevent duplicate submissions where practical.
- Core pages should load within two seconds under the team's defined test conditions, excluding large media downloads.
- Provide useful error messages and retry/resume behaviour for intermittent connectivity.
- Perform regular database backups and test restoration before a real pilot.

4.4 Availability and language

The application should be available during planned school use. English is the initial language unless the team confirms bilingual delivery. Tamil support is an explicit scope decision to resolve before content production.

5. Technical Requirements

5.1 Suggested stack

Layer	Recommended starting choice	Notes
Frontend	React	Responsive component-based interface.
Backend	FastAPI or Node.js/Express	Choose one and freeze it before implementation.
Database	PostgreSQL	Relational data and constraints.
Authentication	Secure cookie/session or carefully implemented token-based auth	Choose based on deployment; enforce server-side roles.
Media	Local storage for development; managed storage for deployment	Keep metadata in the database.
Tools	Git, GitHub, Postman, issue tracker	Use branches, reviews and repeatable tests.

5.2 Main data entities

Entity	Purpose
users	Login identity, role, account status.
students	Student-specific profile, grade/class link.
teachers	Teacher details and class assignments.
schools, classes, class_memberships	Institution and class structure.
subjects, chapters, concepts	Curriculum hierarchy and learning objectives.
intelligence_types	The six supported pathways.
assessment_questions, assessment_options, question_pathway_mappings	Question bank, options, weights and pathway mappings.
assessment_attempts, assessment_responses	Assessment session, responses and timestamps.
intelligence_profiles, profile_scores	Versioned six-pathway profile and ranks.
learning_resources, resource_versions, resource_reviews	Content, pathway tags, approval state and review history.
quizzes, quiz_questions, quiz_attempts, quiz_responses	Concept baseline/post assessments and results.
progress_records, pathway_events	Concept completion and pathway/resource used.
class_test_scores, teacher_feedback, student_feedback	Supplementary assessment and feedback.

5.3 Core relationships

School → Classes → Students

Teachers → Class Assignments → Classes

Subjects → Chapters → Concepts → Learning Resources

Concepts → Quizzes → Quiz Attempts → Students

Students → Assessment Attempts → Assessment Responses

Students → Intelligence Profiles → Profile Scores

Students + Concepts → Progress Records

5.4 Backend services

- Authentication and authorization service.
- Pre-assessment service: question delivery, response persistence, profile scoring.
- Curriculum/content service: browse, tag, approve and recommend resources.
- Quiz service: quiz delivery, scoring, attempts and baseline/post-test association.
- Progress/adaptation service: mastery, learning gain, pathway recommendations and history.
- Reporting service: student progress and class-level summaries.
-

6. Acceptance Criteria and Test Strategy

6.1 MVP acceptance criteria

- A student can register/login and is directed to the preference assessment when no current profile exists.
- The assessment produces six normalized scores, primary and secondary pathways, and a visible profile chart.
- A student can browse a configured sample curriculum and receive approved content associated with the current pathway.
- The student can complete a concept baseline quiz, learning content, and post-quiz; scores and attempts persist.

- Learning gain is calculated correctly for valid pre/post scores and handles a pre-test score of 100 safely.
- Low-performance triggers diagnostic checks before a blended-pathway recommendation; manual override is recorded.
- A teacher can approve content, view assigned students' progress, and enter classroom test marks.
- Unauthorized users cannot access other students' profiles or restricted teacher/admin functions.

6.2 Test categories

Category	Examples
Functional	Registration, assessment, scoring, content selection, quizzes, progress and approval workflow.
Scoring	Six pathway mappings, normalization, missing answers, ties, near-ties, zero denominator.
Adaptation	Good score, low score, incomplete content, insufficient time, manual override, blended pathway.
Security	Role boundaries, direct API access, unauthorized student-record access, session expiry.
Usability/accessibility	Student comprehension, tablet layout, keyboard access, chart labels, media alternatives.
Reliability	Connection loss, duplicate submission, refresh/resume, backup restoration.

6.3 Pilot evaluation

Pilot testing should assess usability, completion, engagement, and concept-level performance. Results must be reported with sample size and limitations. The project must not claim that a small pilot scientifically validates Gardner's theory or proves a universal learning style.

Appendix A — References

Curriculum source: Tamil Nadu school textbooks portal — https://www.tntextbooks.in/p/school-books.html

Questionnaire references used for background and original question design:

- https://psychology.com/tests/multiple-intelligences
- https://alis.alberta.ca/careerinsite/know-yourself/multiple-intelligences-quiz/
- https://www.mentalup.co/blog/multiple-intelligence-test
- https://prismatest.com/en/tests/garner

These online resources are references for question design, not automatic evidence that the resulting MI-Learn assessment is a validated instrument.

Appendix B — Requirement traceability

Project objective	Requirement IDs
Register and assess new student	PRE-F1, PRE-F2, PRE-F5
Six-pathway preference profile	PRE-F3, PRE-F6, PRE-F7, PRE-F8, PRE-F9
Personalized approved curriculum content	LRN-F1, LRN-F2, LRN-F3, LRN-F11
Teacher content review	LRN-F5, LRN-F11
Concept quizzes and learning gain	PST-F1, PST-F2, PST-F4, PST-F5
Adaptive primary/secondary pathway	LRN-F6, PST-F7, PST-F9, PST-F13
Teacher monitoring and class marks	PST-F3, PST-F11
Privacy and access control	Sections 2.3 and 4.2

Appendix C — Decisions to confirm before SRS baseline

ID	Decision	Recommended action
OI-1	Initial language: English or English + Tamil?	Confirm before writing student-facing content.
OI-2	Content authoring for MVP	Use manually authored/teacher-reviewed content; defer AI generation.
OI-3	Parent login in version 1?	Defer unless required by school/mentor.
OI-4	Initial subject and concept set	Choose one class, one subject, and a small number of concepts.
OI-5	Final stack and hosting	Choose one frontend, backend, and database combination.
OI-6	Exact scoring weights and near-tie margin	Finalize and test with sample responses before implementation.
OI-7	Mastery threshold and adaptation rule	Set as configurable pilot parameters; do not claim universal validity.
OI-8	School-visit findings	Add anonymized, relevant observations and requirements.
OI-9	Consent, data retention and deletion process	Confirm with institution before collecting real student data.

Final scope statement

MindBridge is a responsive web application for middle-school students that creates a learning preference profile across six pathways—Linguistic, Logical-Mathematical, Spatial/Visual, Musical, Bodily-Kinesthetic, and Naturalistic. It presents approved curriculum concepts through recommended representations, evaluates learning using concept-level baseline and post-assessments, tracks engagement and performance separately, and can recommend a secondary or blended pathway when evidence indicates that adaptation may help. The system supports learning decisions; it does not diagnose or permanently classify student intelligence.